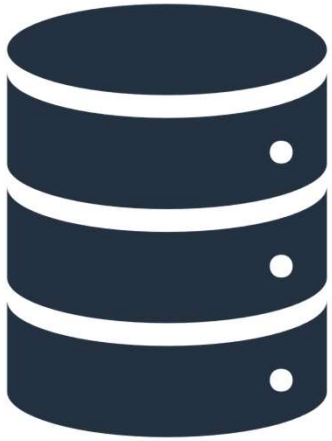
DATA ENGINEERING:
BIOMASS PRE-PROCESSING DATABASE



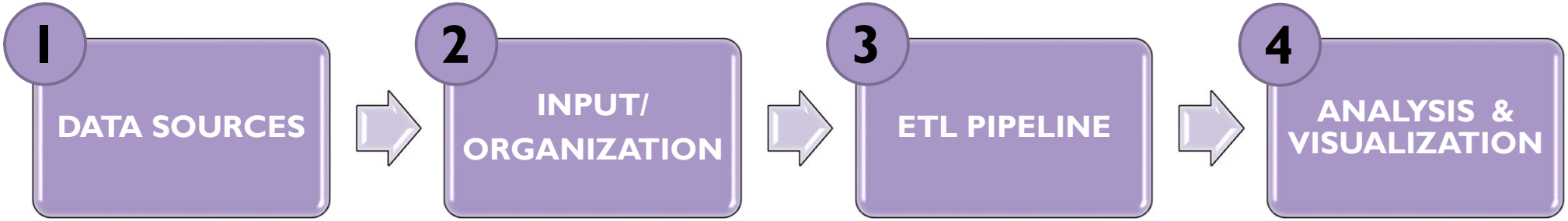
Amanda Hydar

Research Funded by USDA Sun Grant Initiative

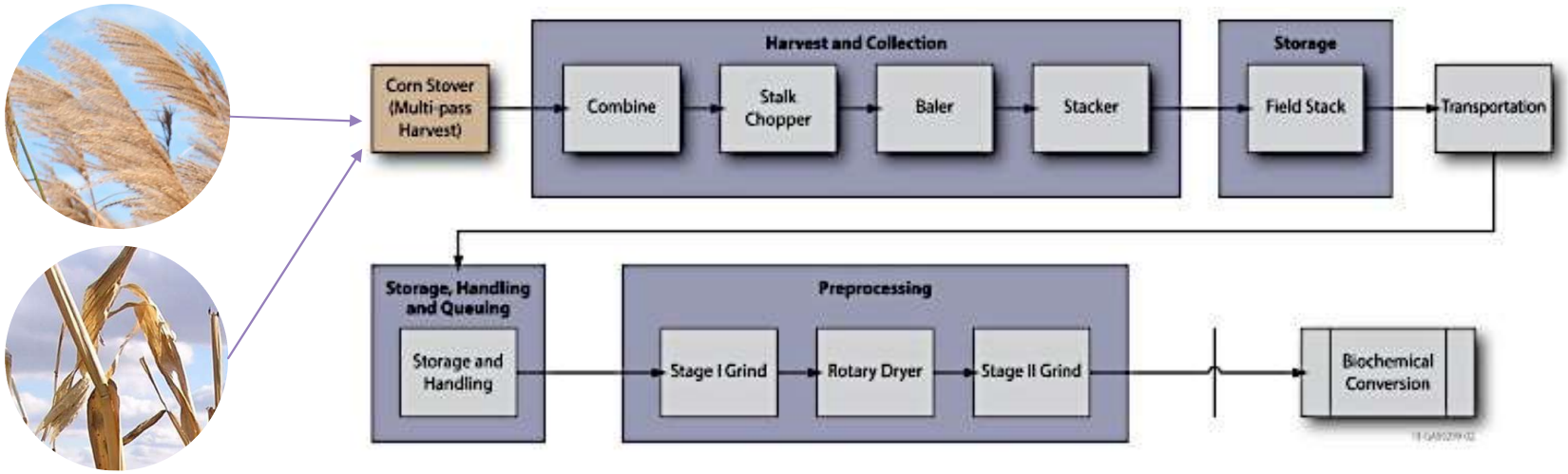
THE OBJECTIVE:

Create a database of literature on cutting edge technologies and input process data for modeling

THE SOLUTION:



BACKGROUND: THE BIOMASS-TO-BIOFUEL SUPPLY CHAIN



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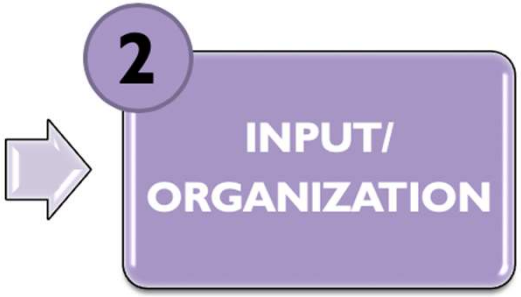
DATA SOURCES

BIOENERGY
KNOWLEDGE DISCOVERY FRAMEWORK
KDF
U.S. DEPARTMENT OF ENERGY

- CSV, TXT, Excel files from participating laboratories & universities
- Manual data entry
- Processing Demonstration Unit at INL



- Data characterization
- Process inputs/outputs
- Critical requirements

BIOMASS DATA

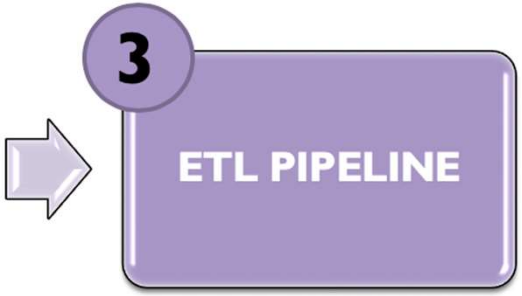
Variable Name	Values or Data Types	Definition	Importance	Notes
biomass_type	1 = Corn Stover 2 = Switchgrass 3 = Miscanthus 4 = Other	Type of biomass	5 Cannot Be Missing	
particle_size	numeric	Particle size	4 Absolutely Necessary	Units: mm
aspect_ratio	numeric	Aspect Ratio	3 Necessary for Multiple Analyses	Format all ratios from xx:yy to a decimal fraction

EQUIPMENT DATA

Variable Name	Values or Data Types	Definition	Importance	Notes
equipment_type	1 = Grinder 2 = Conveyor 3 = Metering Bin 4 = Pellet Mill 5 = Feeder 6 = Reactor	Type of equipment (biomass INPUT)	5 Cannot Be Missing	Format make/model info for general type of equipment
process_stage	numeric	Stage of	3 Necessary for Multiple Analyses	Refer to proposed preprocessing and reactor feeding system [3]
moisture_content	numeric	Moisture content of incoming biomass	3 Necessary for Multiple Analyses	Percentage as a 0.XX decimal

MODELING TECHNIQUES

Variable Name	Values or Data Types	Definition	Importance	Notes
author	list	Author of published paper	5 Cannot Be Missing	Include all authors as a list
year	numeric	Year published	5 Cannot Be Missing	
model_tech	string	Modeling technique	5 Cannot Be Missing	Refer to paper matrix for string [9]



UPLOAD TRACKER

Original File Name	*File Source*	*Clean Data File Name*	*Upload Date YYYY-MM-DD*
raw_biomass_props	INL Report [1]	biomass_props	2019-01-01
raw_paper_matrix	Gathered data [2]	paper_matrix	2019-01-01

Check for missing critical data

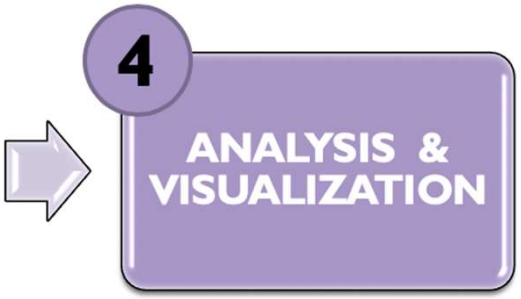
- Filter rows with missing critical data points
- Keep all raw files untouched

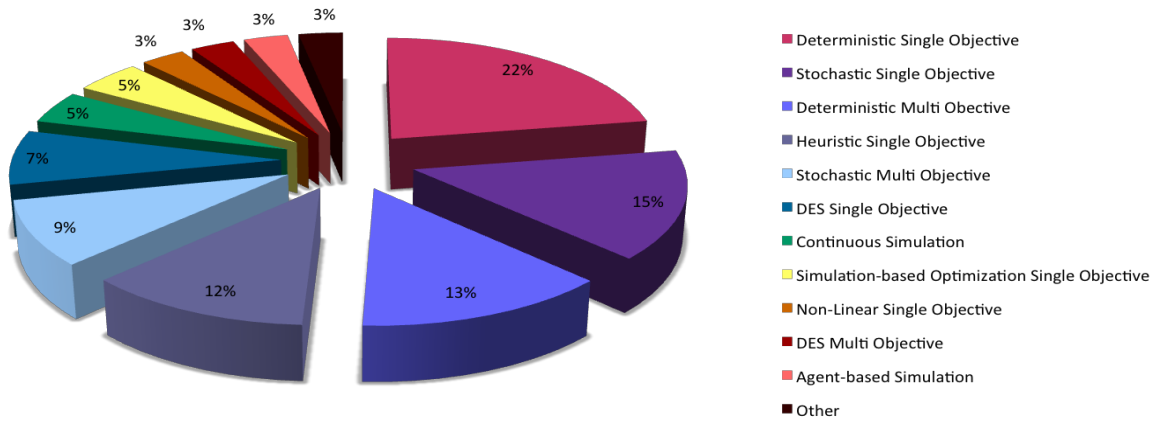
Quality Assurance Check

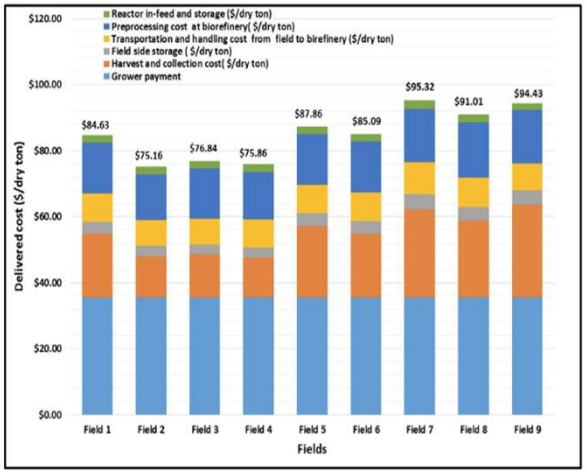
- Detailed procedure to check upper/lower bounds
- Additional checks for unforeseen anomalies

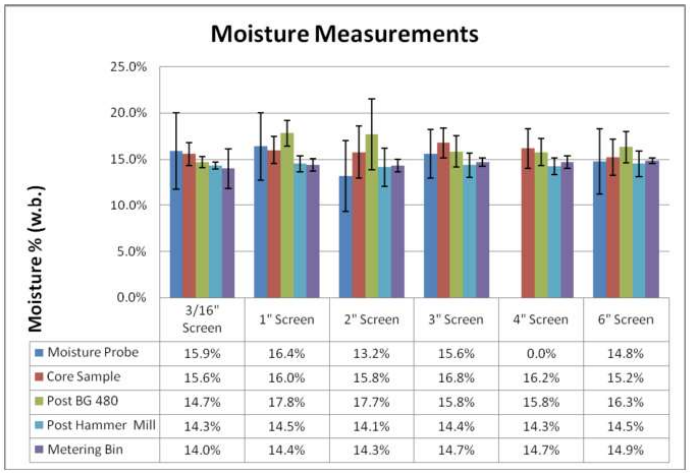
Structured Import

- All uploads follow import procedure
- Upload tracker to record all changes









	3/16" Screen	1" Screen	2" Screen	3" Screen	4" Screen	6" Screen
Moisture Probe	15.9%	16.4%	13.2%	15.6%	0.0%	14.8%
Core Sample	15.6%	16.0%	15.8%	16.8%	16.2%	15.2%
Post BG 480	14.7%	17.8%	17.7%	15.8%	15.8%	16.3%
Post Hammer Mill	14.3%	14.5%	14.1%	14.4%	14.3%	14.5%
Metering Bin	14.0%	14.4%	14.3%	14.7%	14.7%	14.9%

Minitab, R, Python, SQL, Tableau

- Centralized & clean data simplifies analysis

Future Areas of research

- Insights can direct research iteratively